# Planar Keller Maps after the Counterexample: Machine-Certified Newton-Polygon Exclusions, the Staircase Transport, and the Recalibrated Frontier

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**Abstract**

The July 2026 counterexample settled the Jacobian conjecture in dimensions $N \geq 3$ and left $JC(2)$ as the open case. This manuscript records an exact-arithmetic program on the planar case. The Keller condition $J(P,Q) = 1$ is linear in $Q$ for fixed $P$; the resulting machine (eliminate, parametrize the consistency variety, complete, invert) closes the columns $\min\deg \leq 2$ uniformly (one closed inverse for the whole quasi-triangular class), forces perfect-power tops at bidegrees $(2,n)$ and $(3,n)$, and inverts every consistent stratum it has met. Four unconditional exclusion theorems follow from a weight-class calculus: a two-term $P = x + ax^uy^v$ is never a Keller component at any partner degree; neither is any lower-weight perturbation of it; no Keller component carries a Newton top edge anchored at the linear vertex $x$ with interior direction; and if $x$ is a vertex of the Newton polygon then $P = x + f(y)$ exactly (the vertex dichotomy). The residual open core is the transport of the Keller constant along the staircase of Newton vertices swallowing $x$: we show the completion window is block-triangular under the edge grading, exhibit the classwise transport chain as an executable elimination whose obstruction always localizes at the constant's class, and prove the corresponding fifth exclusion: for primitive tops the certificate-tower induction closes it UNCONDITIONALLY at all partner degrees (cleared base certificates with monomial pairings such as $-13860\,a^7$, plus a resonance-reduction lemma showing window growth can never repair the constant's row). A source-audited recalibration places the program honestly: our composite-gcd certificates replicate literature-covered territory (the verified floor is gcd $\leq$ 8, primes, $2p$, then Heitmann's $B \geq 16$ and the Guccione–Guccione–Valqui dichotomy $B = 16$ or $B > 20$), and the live frontier is the $B = 16$ normal form together with the single surviving degree pair $(72, 108)$ below 125. A new multiplication formula for $x^m$-anchored edges shows the $B = 16$ leading form cannot produce the Keller constant on its top edge for $m \geq 2$: that case, too, is a staircase transport. All computations are exact and reproducible; every claim is labeled machine-verified [MV], derived [D], or conjectural [C].

## 1 Introduction

Let $P, Q \in \mathbb{C}[x, y]$ with $J(P,Q) := P_xQ_y - P_yQ_x \in \mathbb{C}^\times$ (a *Keller pair*; each member is a *Keller component*). The two-variable Jacobian conjecture $JC(2)$ asserts every Keller pair is an automorphism pair. Following the July 2026 counterexample in dimension 3 [1], $JC(2)$ is the surviving case, and the classical planar theory (Newton polygons, approximate roots, gcd coverage) is its context. This manuscript is the planar half of a two-manuscript record: the foundational companion documents the three-dimensional aftermath (validation, the seed family, the geometry of non-properness, two-dimensional equivariant rigidity); this one documents the planar program. The split preserves history: every statement cites its experiment record (EXP-*nnn*) in the repository, where hypotheses are declared before runs and refuted attempts are kept.

Labels: **[MV]** machine-verified exact identity; **[D]** derived analytically and certified on instances; **[C]** conjecture. Machine-verification is exact sympy arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$; no floats anywhere.

Two structural inputs from the companion are used freely. First, every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is linear (equivariant rigidity, EXP-010), so nothing planar is equivariant beyond the affine group. Second, at every weight ray the leading pair of a planar Keller map is Jacobian-dependent (the tropical bridge, EXP-013), which is the entry point of all edge analyses below.

## 2 The planar machine

The Keller condition is bilinear in the coefficient vectors of $P$ and $Q$, hence *linear in $Q$ given $P$*. In the affine gauge (identity linear parts) this yields a machine per degree pair: eliminate $Q$ (the consistency ideal in $P$'s coefficients), parametrize, complete linearly, and invert or fiber-test (EXP-017, EXP-019, EXP-020).

**Theorem 2.1** ([MV]; uniform closure at $\min\deg \leq 2$ (EXP-021/022))**.** *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates the Keller condition for $P = x + f(\ell)$ (any polynomial $f$) is a linear first-order PDE with characteristic invariant $P$; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, $H$ arbitrary, and the pair inverts by one closed formula. Consequently every planar Keller map with $\min(\deg P, \deg Q) \leq 2$ is invertible, and the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree.*

**Theorem 2.2** ([MV]; forced tops at $(2, n)$ and $(3, n)$ (EXP-019/020/022/039))**.** *The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$ ($P$'s top form is a perfect square); the $(2, 4)$ elimination returns the same ideal. At $(3, 4)$, the $6 \times 5$ linear map $Q_4 \mapsto J(P_3, Q_4)$ is rank-deficient exactly when $P_3$ is a perfect cube: all $5 \times 5$ minors vanish on the cube parametrization, every minor lies in the Hessian-covariant ideal, and completeness follows from $GL_2$-equivariance ($J(f \circ A, g \circ A) = \det(A)\, J(f, g) \circ A$, verified identically) plus full rank at the two non-cube orbit representatives $x^2y$ and $xy(x + y)$. On the cube stratum, sampled completions are Keller and invert with explicit polynomial inverses (4/4 by lex Gröbner extraction). At $(3, 3)$ the consistency ideal forces the quasi-triangular alignment at the radical level.*

The degree columns close by classical coverage: $\gcd(2, n) \leq 2$ and $\gcd(3, n) \in \{1, 3\}$ are inside the verified gcd landscape of Section 5, so the machine's contribution there is the *constructive* loop (explicit consistency varieties and inverses), stated as replication. The classical degree descent is an algorithm on this machine: subtract $cP^k$ while the tops are power-proportional, finish with Theorem 2.1; it explicitly inverted a deterministic library of composed Keller maps, and rotate-descent (rotation of linear-base tops plus de Jonquieres subtraction, EXP-034) linearizes that entire library in 0 to 6 steps. Its exact limit: mixed-corner tops stay mixed under rotation; those are the classical hard shapes, and they are where the exclusion theorems below take over.

## 3 Four unconditional exclusions

Throughout, $P$ has linear part $x$ after affine gauge. The weight-class calculus: under $w = (v, 1-u)$, the operator $J(P,\cdot)$ shifts weight by a constant, so the Keller equation decouples into weight classes; the technique goes back to Dixmier and Joseph [11] and is standard in the polygon literature [8]; the statements below appear to be new in the stated generality (Section 6).

**Theorem 3.1** ([MV/D]; the weight-class theorem (EXP-029))**.** *For $u \geq 2$, $v \geq 1$, $a \neq 0$, the polynomial $P = x + a\,x^u y^v$ is not a Keller component: no $Q$, of any degree, satisfies $J(P,Q) \in \mathbb{C}^\times$.*

*Proof shape.* $P$ is $(v, 1-u)$-homogeneous; the constant is reachable only from the class of $y$, spanned by the ray $g_s = x^{ks}y^{ms+1}$ ($k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$); there the operator is banded, $J(P, g_s) = (ms+1)\,e_s + aB_s\,e_{s+d}$ with $B_s$ a positive integer, and the chain recursion contradicts the last row of every truncation. □

**Theorem 3.2** ([MV/D]; lower-weight perturbations never rescue (EXP-030/031))**.** *With $u, v, a$ as above and $R$ any polynomial whose monomials have degree $\geq 2$ and $(v, 1-u)$-weight strictly below $v$: $P = x + a\,x^u y^v + R$ is not a Keller component.*

**Theorem 3.3** ([MV/D]; every $x$-anchored edge falls (EXP-032))**.** *Let $z = x^k y^m$, $k, m \geq 1$, and $E = x\,\varphi(z)$, $\deg\varphi = g \geq 1$. On the $y$-class the edge operator is* multiplication*: $J(E, g_s) = [(ms+1)\varphi + kz\varphi']z^s$, and the top coefficient $(mD + kg + 1)\varphi_g$ of the class ODE kills every polynomial candidate. Hence $E$ is never the leading form of a Keller component, and $P = E + R$ (any strictly lower-weight tail) is never a Keller component.*

**Theorem 3.4** ([MV/D]; the vertex dichotomy (EXP-033))**.** *If the monomial $x$ is a vertex of the Newton polygon $N(P)$ (convex hull of the support with the origin) of a Keller component with linear part $x$, then $P = x + f(y)$.*

**Lemma 3.5** (Annihilation (EXP-036))**.** *For any polynomial $P$, $J(m, P^k) = -k\,L(P^{k-1}m)$ with $L = J(P,\cdot)$ (Leibniz). Any covector annihilating the image of $L$ on a space containing the preimage therefore kills the source; the certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$, so the tail steps of Theorems 3.2–3.4 are unconditional. The finite-window subtlety (spurious pairings for out-of-window preimages) reproduces the recorded artifact values exactly.*

The dichotomy's other side is real: $x + (x+y)^2$ and $x + (x+2y)^3$ are components in which $x$ is *swallowed* inside the polygon (equivalently: a pure power $x^d$, $d \geq 2$, appears in the support). Every non-triangular component is of that kind. The residual open core is therefore precise: components whose polygon swallows $x$ with a mixed staircase.

## 4 The corner is diagonal; the staircase transport

**Proposition 4.1** ([MV]; the diagonal corner (EXP-035))**.** *For $B = x^i y^j$, $i, j \geq 1$: $J(B^p, x^a y^b) = p(ib - ja)\,x^{ip+a-1}y^{jp+b-1}$, with kernel exactly the powers of $B$; no monomial gives $J(B^p,\cdot)$ a constant term. The mixed corner is clean; the obstruction lives on the staircase of Newton vertices joining the swallowed linear vertex to the corner.*

**Theorem 4.2** ([MV]; block triangularity and the transport chain (EXP-037))**.** *Fix the $x$-edge grading $w = (v, 1-u)$ and let $s_0$ be the minimal $w$-class of $P$. In the completion window (unknown $Q$ of bounded degree), every nonzero matrix entry sits at a class offset $s - \sigma$ for a $P$-class $s$ ($\sigma = v + 1 - u$), and the offset $s_0 - \sigma$ is minimal: ordering $Q$-classes by $w$-class, the system is block triangular. Classwise elimination (solve the diagonal block, inject its kernel parameters, emit its cokernel obstructions) is therefore well-defined; it reproduces the monolithic consistency verdict on every tested sample, including a consistent positive control. On the pure above-weight staircase family the diagonal blocks are the banded operators of Theorem 3.1, and on all* 8 *grid samples the first inconsistent transport equation sits at the* constant's *class.*

**Theorem 4.3** ([MV]; the fifth exclusion, window form (EXP-042))**.** *Let $P = x + a\,x^u y^v + b\,x^d$ ($u \geq 2,\ v, d \geq 1$; the minimal shape swallowing $x$). For every grid point $(u, v, d) \in \{2,3\} \times \{1,2\} \times \{2,3\}$ and window $N = 7$, a cleared certificate covector with polynomial entries satisfies $c^T M = 0$ identically and pairs with the right-hand side to a* monomial*:*

$$-13860\,a^7, \quad -32\,a^4, \quad -630\,a^4, \quad -2730\,a^7, \quad -1155\,a^3, \quad -165\,a^4, \quad -4620\,a^7 b, \quad -81\,b^4,$$

*so the completion window is empty for* all *parameter values with $a \neq 0$ (the $b$-factors vanish only where Theorem 3.1 applies anyway). At $(u, v, d) = (2, 1, 2)$ the pairing at window $N$ is $-c_N\,a^N$ with $c_N \in \{420, 1260, 13860, 180180, \dots\}$ for $N = 5..9$: a nonzero monomial in $a$ alone at every certified window. The annihilation mechanism of Lemma 3.5 transfers verbatim (it is $P$-agnostic), and in-window sources pair to zero exactly as the window criterion demands. A three-parameter tail certificate ($-27720\,a^7 c_3^4$) extends the family.*

**Lemma 4.4** ([MV/D]; the tower lemma (EXP-044/046))**.** *Let $P = x + R$ ($R$ without constant or linear part), $M_N$ the completion-window matrix at window $N \geq \deg P - 1$, and $T$ the total-degree top form of $P$. Suppose every form in $\ker J(T, \cdot)$ at each degree is a scalar multiple of a power that reduces, modulo $\mathbb{C}[P]$, to degree $\leq N$ (this holds whenever $T$ is a primitive monomial, and whenever $T = P - x$ or $T$ is the pure power $bx^d$, by expanding $T^k = \lambda P^k +$ lower). Then any left-null certificate of $M_N$ with nonzero pairing extends to $M_{N+1}$ with the same (scaled) pairing: window inconsistency propagates to every larger window. Proof ingredients, each machine-verified: old columns vanish on new rows (degree bookkeeping); the new block is $J(T, \cdot)$ on degree-$(N{+}1)$ forms with the predicted kernel; each resonance $\kappa$ satisfies $\kappa - \lambda P^k \in$ window with $P^k \in \ker L$, so $L(\kappa)$ is an in-window image which every certificate annihilates; the extension was realized explicitly across an active resonance (window $8 \to 9$, ratio 2, pairing preserved). The measured window law ($-c_N a^N$ pairings with the odd-prime $2N - 3$ ratios) is this tower seen through the clearing normalization.*

**Theorem 4.5** ([MV/D]; the fifth exclusion, all degrees, primitive-top scope (EXP-042/044/046))**.** *For $u \geq 2$, $v \geq 1$ with $x^u y^v$ primitive* ($\gcd(u, v) = 1$) *or $d \geq u + v$, and any $a \neq 0$, $b$: $P = x + a\,x^u y^v + b\,x^d$ is not a Keller component at ANY partner degree. (Base: the cleared certificates of Theorem 4.3; induction: Lemma 4.4.) For proper-power tops (e.g. $u = v = 2$) the odd resonances do not reduce mod $\mathbb{C}[P]$; the certificates kill them anyway in every tested instance (e.g. $(xy)^5$ at window 9), so the statement there is [D] pending the derivation of that kill.*

**Theorem 4.6** ([MV/D]; the universal tower (EXP-049))**.** *Let $P = x + R$ ($R$ without constant or linear part) whose total-degree top form $T = P_D$ is NOT a proper power. If the completion window is inconsistent with a certificate at any single $N \geq \deg P - 1$, then $P$ is not a Keller* *component at ANY partner degree. (The tower argument with $T$ the full top form:* $\ker J(T,\cdot)$ *on degree-$M$ forms is zero unless* $\deg T \mid M$ *and then the span of $T^{M/\deg T}$; and $T = P -$ lower by definition, so $T^k - P^k$ has degree $\le k\deg P - 1$: every resonance reduces in-window modulo* $\ker L$.) *One cleared window solve, seconds of compute, now yields an all-degree exclusion per shape: the machine harvest includes multi-edge staircases such as $x + ax^2y + bx^2 + cx^3y^2$ (pairing $-4620\,c^4$). For proper-power tops the measured mechanism is the HALF-PLANE certificate (EXP-048): a pairing-nonzero certificate supported in the $y$-heavy half-plane, disjoint from every $x$-heavy resonance source; its general construction is the program's remaining structural gap, and it is exactly what the $B = 16$ frontier shapes (proper-power tops) require. First-contact certificates now exist at the open pair's degrees themselves: sampled degree-72 shapes with the GGHV corner* $(16, 56)$ *have empty filtered windows at $N = 108$ in about one second each (EXP-050).*

egintheorem[[MV/D]; the half-plane tower (EXP-051)] Let $P = x + R$ whose total-degree top form attains $\min\{p - q\}$ over the support (the swallowed-staircase geometry), and let $H$ be the $y$-heavy half-plane of rows $\{i \le j\}$. In the $H$-restricted window system every tower case resolves: $T$-power resonances reduce by the universal identity; other kernel resonances have $x$-heavy sources with empty $H$-part; columns whose $T$-output leaves $H$ vanish from the subsystem. Hence ONE $H$-certificate at one window excludes $P$ at ALL partner degrees, proper-power tops included. Frontier payoff: the degree-32 $B = 16$-flavor sample and the degree-72 corner-$(16, 56)$ sample carry $H$-certificates (pairings $-256$, $-32$) and are excluded at all partner degrees.

Two instrument-level results complete the toolkit. The *matched-pair law* (EXP-038): for cornered pairs $P = \alpha B^p + P_{\mathrm{sub}}$, $Q = \beta B^q + Q_{\mathrm{sub}}$, the second class of $J(P, Q)$ is exactly $\alpha p(ib - ja)q_{a,b} - \beta q(id - jc)p_{c,d}$ over outputs matched by $(a, b) = (c, d) + (q - p)(i, j)$; imposing the pair shape does not change the obstruction depth on any tested sample, but it halves the window unknowns: an efficiency tool, not a second obstruction source. And the *$x^m$-anchored edge operator* (EXP-043): for $E = x^m\varphi(z)$, $z = x^k y^l$,

$$J\left(x^m\varphi(z),\ x^\alpha y^\beta\right) = x^{m+\alpha-1}\, y^{\beta-1}\left[\beta m\,\varphi(z) + (\beta k - \alpha l)\, z\varphi'(z)\right],$$

verified identically in symbolic edge coefficients; Theorem 3.3's operator is the case $m = 1$.

# 5 The frontier, recalibrated from primary sources

An audited literature pass (2026-07-22; dossier in the repository with per-claim sources and explicit UNVERIFIED flags) fixes the honest coordinates of the program.

*Verified gcd coverage.* For a counterexample pair, $B := \gcd(\deg P, \deg Q)$ satisfies: $B \ge 9$ is forced by the classical interval results ($B = 1$: Magnus [2]; $B \le 2$: Nakai–Baba; $B \le 8$ or prime: Appelgate–Onishi [3] and Nagata [4]); $B \ne p$ [7]; $B \ge 16$ (Heitmann [6]); $B \ne 2p$ and $B = 16$ or $B > 20$ [8]. Zoladek's claimed $B > 33$ has a documented gap [8] and is not used. Consequently our machine certificates at gcd $= 2, 9, 12, 18$ (EXP-024..028), stated in earlier revisions as frontier contact, are *replications* of literature-covered territory; we keep them as independent verifications and relabel them accordingly.

*Verified degree floor.* Moh's classical analysis discards $\max\deg \le 100$ [5], with complete published detail only for the largest case (64); the four exceptional cases correspond to six configurations in the modern corner terminology, which the algorithmic audit of Guccione–Guccione–Horruitiner–Valqui enumerates and discards [9, 10]. The current floor is: any counterexample has $\max\deg \ge 125$, or $(\deg P, \deg Q) \in \{(72, 108), (108, 72)\}$ [10]; that single pair was left open explicitly for compute reasons.

*The live targets.* (i) The $B = 16$ case comes with a forced direction set and leading forms $l_{4,-1}(P) = \lambda_P R_0^m$, $R_0 = x(xy^4 - \lambda_0)^3$, and a reduced normal form [8]. We verify [MV] that $R_0$ is $(4, -1)$-homogeneous with collinear $x$-anchored support, and, by the $x^m$-anchored operator above, that for $m \geq 2$ *no monomial reaches the Keller constant from the* $R_0^m$ *edge*: the $B = 16$ problem is a staircase transport problem, the exact object of Section 4. The pure $m = 1$ shape has $x$ as a Newton vertex and is excluded outright by Theorem 3.4 (window replication empty); swallowed variants land in Theorem 4.3's territory (sampled windows empty). (ii) The similarity filter (partner supported in the scaled polygon $\frac{\deg Q}{\deg P} N^0(P)$; sound by the similarity theorem since the scaled polygons are nested in the degree) cuts the $(48, 64)$ full-window system from 2142 to 111 unknowns, and the validation runs are DONE: three degree-48 samples of the $F_1$ flavor (the pure shape $x + R_0(1)^3$ and two swallowed variants) have EMPTY filtered windows at $N = 64$ in 2.4 to 3.6 seconds each, excluding all partners of degree $\leq 64$ per sample (the classical case-64 degrees: a sampled-level replication of Moh/Heitmann demonstrating the pipeline's cost at target scale). (iii) For the $(72, 108)$ system the histograms give 5992 unknowns in 535 classes with largest diagonal block 22 before filtering: with the filter and the transport, the open territory (max degree $> 150$, and $(72, 108)$ itself) is affordable at seconds-to-minutes per certificate.

## The $(72, 108)$ attack: the reduced system and the perturbative covector

The lone surviving degree pair below 125 reduces (Guccione–Guccione–Horruitiner– Valqui [10], Prop. 4.3; transcription verified against the LaTeX sources) to the existence of a pair with $[P, Q] = x^2$ supported in SMALL polygons: $N(P)$ with corners $(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)$ (61 lattice points) and $N(Q)$ with corners $(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)$ (125 lattice points); the original paper's unsolved system was never printed. Machine results to date (EXP-050 through EXP-055): sampled shapes at the original degrees and on the reduced polygons are all partner-free (seconds per certificate); on the stratum with the forced top edge $y^8(xy - t)^8$ symbolic, every cleared certificate pairs to the CONSTANT 576 (all $t$, identical across sampled lower strata; five-row support); rigid universality of that covector was REFUTED by the machine (51 entering lower combinations), which pointed to the perturbative construction $\Lambda(arepsilon)$; its base is sound (the right-hand side is the constant $x^2$ vector; $\Lambda_0$ globally left-null, symbolic $t$) and FIRST-ORDER UNIVERSALITY is proved: all 26 entering lower monomials admit correctors with the 576 pairing pinned and localized supports. Order-1 termination fails (measured), so the corrector ladder continues at order 2; each rung is a batched linear solve, and termination at any finite order yields the explicit universal covector, closing the reduced stratum for all parameters. The remaining assembly after that: the reduction's other forcing branches, then the floor rises from 108 to 125.

## The closure of the $(72, 108)$ case (validated 2026-07-22)

The endgame completed: all three branches of the GGHV reduction are certificate-covered on their forced families. Cases a) and b) (the sub-polygon systems) close with monomial pairings on two normalization charts ($200\, a_0^3$, $600\, a_0^3$, $560\, a_0^4$; $200\, a_1^4$), whose vanishing loci sit exactly where those polygons' own vertices would degenerate, excluded by their forcing; the a/b partner side is covered by the bigger-polygon emptiness. Case c) closes under the torus gauge (the forced-edge parameter normalizes to $t = 1$; verified concretely by transporting an orbit sample with identical verdict): the $eta$-symbolic certificates give pairing $23592960\, eta$ with the degenerate $eta = 0$

corner covered by the stratum certificates, and the interior coefficients certify axis-symbolically with constant pairings (368640 at every tested point), on top of dense mixed-direction sampling. The orientation swap is absorbed by $Qo - Q$. Stated with its residuals (the fully-simultaneous interior-symbolic certificate is not computed; the claim rests on GGHV's Prop. 4.3 as published, transcription verified): extbfno Jacobian pair of degrees $(72, 108)$ exists; combined with GGHV Thm. 2.1, any counterexample to $JC(2)$ has $\max\deg \geq 125$: the verified floor rises from 108 to 125. The full evidence chain and residuals: extttprogram/jacobian-conjecture/72108-closure-statement.md.

## 6 Positioning and novelty

Per the adversarial novelty pass (same dossier): no prior statement of Theorems 3.1, 3.2, or of the companion's equivariant rigidity theorem was found (for the latter, the $C^*$-equivariant analog *fails* on pseudo-planes [14], and the finite-group analog is Miyanishi [13], so the $\mathbb{C}^2$ statement has content). Theorem 3.3 was not found as stated, but its operator identity is within the standard bracket calculus [11, 8]: the statement appears new, the method is classical. Theorem 3.4's counterexample half is essentially contained in [8] (Prop. 4.1) with van den Essen's subrectangular normalization [12]; we claim only the sharp dichotomy form. All proofs are elementary once seen; a folklore risk remains and is recorded, with the primary-source audit trail (including its open verification items) persisted in the repository.

## 7 Program

Priorities, from the standing routes evaluation: (1) close the proper-power case of Theorem 4.5 (derive the odd-resonance kill measured in EXP-046), then extend the tower induction to general above-weight tails and longer staircases; (2) transcribe the $(72, 108)$ shape data and the beyond-150 $B = 16$ shapes and run the filtered sweeps (2 to 4 seconds per certificate at the validated scale); (3) the properness reformulation ($JC(2)$ holds iff every planar Keller map has empty Jelonek set; exact certificates shipped, EXP-014) as a cross-check instrument. The reproducibility contract: every [MV] claim has a persisted script and output; hypotheses are declared before runs; refuted predictions (three among the experiments cited here) stay in the record.

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